

This is known as the **mass number**. This is the number of **protons and neutrons** added together

This is known as the **atomic number** or **proton number.** This is the number of protons in

Number of neutrons = mass number – atomic number

The number of **electrons** is the same as the number of protons in an atom.

Task 1

Using your periodic tables fill in the table below with the correct numbers.

Clue: Some of the numbers will be the same. If you can work out which numbers they are, it will make filling in the table easier.

Element	Atomic number	Mass number	Number of protons	Number of neutrons	Number of electrons
Hydrogen					
Helium					
Lithium					
Beryllium					
Boron					
Carbon					
Nitrogen					
Oxygen					
Fluorine					
Neon					
Sodium					
Magnesium					
Aluminium					
Silicon					
Phosphorus					
Sulfur					
Chlorine					
Argon					
Potassium					
Calcium					

Task 2

Answer the following questions.

1. What is the pattern shown by the atomic numbers of the first 20 elements?

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2. Three of the numbers on the table are always the same for each element. Which three numbers are they?

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3. There is only one element that has a mass number that is not higher than its atomic number. Which element is it? Use the table to answer this question.

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4. Which element has the lowest mass? .....

5. Name the elements from the table with the same number of protons and neutrons.

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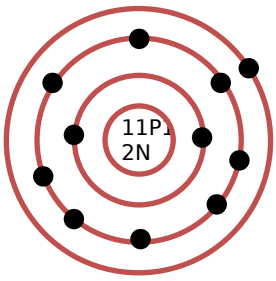
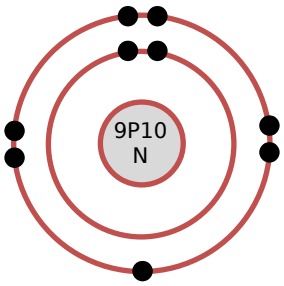
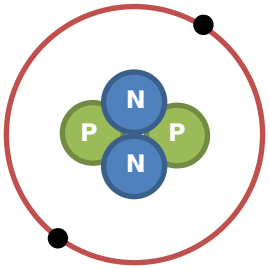
6. Look at the elements lithium, sodium, potassium and rubidium in the periodic table. What happens to their atomic numbers as you look down the column?

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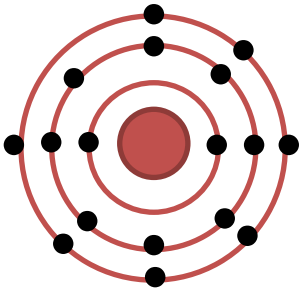
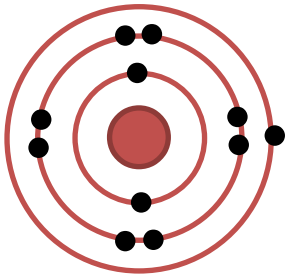
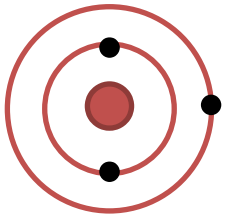
7. Look at beryllium, magnesium, calcium and strontium. Do their atomic numbers change in the same way as the elements in question 3?

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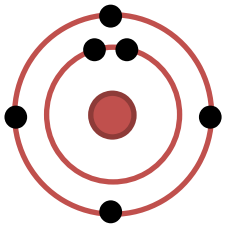
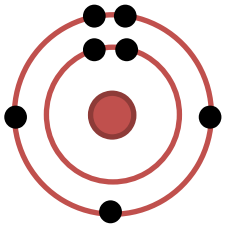
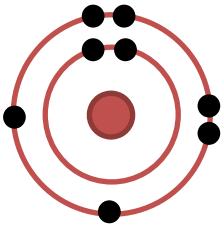
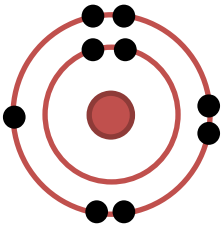
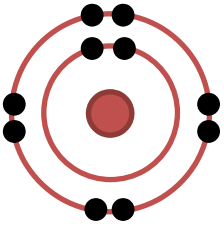
8. Name these elements (P = protons, N = neutrons, small dots = electrons). Explain how you named these elements.

Element A	Element B	Element C
		

9. Name these elements using the diagrams. Explain how you named these elements. (Hint: The dots on circles are electrons. The number of protons and neutrons are not shown.)

Element A	Element B	Element C
		

10. How many neutrons do these atoms have? Explain how you worked this out.

Element A	Element B	Element C	Element D	Element E
				
Number of neutrons				

Answers

Element	Atomic number	Mass number	Number of protons	Number of neutrons	Number of electrons
Hydrogen	**1**	**1**	**1**	**0**	**1**
Helium	**2**	**4**	**2**	**2**	**2**
Lithium	**3**	**7**	**3**	**4**	**3**
Beryllium	**4**	**9**	**4**	**5**	**4**
Boron	**5**	**11**	**5**	**6**	**5**
Carbon	**6**	**12**	**6**	**6**	**6**
Nitrogen	**7**	**14**	**7**	**7**	**7**
Oxygen	**8**	**16**	**8**	**8**	**8**
Fluorine	**9**	**19**	**9**	**10**	**9**
Neon	**10**	**20**	**10**	**10**	**10**
Sodium	**11**	**23**	**11**	**12**	**11**
Magnesium	**12**	**24**	**12**	**12**	**12**
Aluminium	**13**	**27**	**13**	**14**	**13**
Silicon	**14**	**28**	**14**	**14**	**14**
Phosphorus	**15**	**31**	**15**	**16**	**15**
Sulfur	**16**	**32**	**16**	**16**	**16**
Chlorine	**17**	**35**	**17**	**18**	**17**
Argon	**18**	**40**	**18**	**22**	**18**
Potassium	**19**	**39**	**19**	**20**	**19**
Calcium	**20**	**40**	**20**	**20**	**20**

1. They go up in 1s.
2. Atomic number, number of protons and number of electrons.
3. Hydrogen – mass number is 1 which is the same as its atomic number.
4. Hydrogen.
5. Helium, carbon, nitrogen, oxygen, neon, magnesium, silicon, sulfur, calcium.
6. They increase by 8.
7. Yes.
8. a)sodium, b) fluorine, c) helium

   Add the neutrons and protons together to work out the mass number.
   Or match the number of electrons (or protons) to the atomic number.

9. a)chlorine b) sodium c) lithium

   The number of electrons equals the number of protons (which is the same as the atomic number). Find the element with that atomic number.

10. a) 6 neutrons b) 7 neutrons c) 8 neutrons d) 10 neutrons e) 10 neutrons

    Look at the number of electrons. The number of electrons equals the number of protons, then look for the elements with the same atomic number as the number of electrons. Then take the mass number and subtract the atomic number from it.